

A. A.A The smallest class of f is C^2 .

$$\text{A.B } f(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ x^{k+1} & \text{if } x > 0 \end{cases}.$$

A.C First, we show that each of the component functions of f are themselves smooth. Let $h(x) = 0$, which is constant. Therefore, $h^{(k)}(x) = 0$, and as such is smooth. Let $g(x) = e^{-x^{-1}}$. It is an exponential, and therefore is smooth. We need only now consider $x = 0$, where we show that $\lim_{x \rightarrow 0^-} f^{(k)}(x) = \lim_{x \rightarrow 0^+} f^{(k)}(x)$ for all $k \geq 0$. The k th derivative of $g(x)$ is

$$\frac{d^{(k)}}{dx^{(k)}}(-x^{-1})e^{-x^{-1}}. \quad (1)$$

The k th derivative of $-x^{-1}$ is in the form $\pm \frac{1}{x^k}$, and therefore the k th derivative of $g(x)$ is $\pm \frac{e^{-x^{-1}}}{x^k}$. We assume without proof that $\lim_{x \rightarrow 0^+} \frac{e^{-x^{-1}}}{x^k} = 0$. We have previously shown that $h^{(k)}(x) = 0$, and $\lim_{x \rightarrow 0^-} 0 = 0$. Therefore, we have shown f to be smooth, and we are done.

A.D f is not analytic.

Proof. We show that f is not analytic by showing that its Taylor-Maclaurin series does not converge to f on all of $\mathbb{R}$. The Taylor-Maclaurin series for $e^{-x^{-1}}$ is $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!x^n}$. When $x = 0$, this sum does not exist, and as a result the Taylor-Maclaurin series of f does not converge to f on all of $\mathbb{R}$. Thus, we are done. $\square$

B. B.A For a positive-definite, symmetric, $m \times m$ matrix G , there exists a $\sqrt{G}$ such that $(\sqrt{G})^2 = G$.

Proof. Let G be a symmetric, positive-definite, $m \times m$ matrix. By the spectral theorem, we know that G has m orthogonal (and thus linearly independent) eigenvectors, and a corresponding m real eigenvalues. Furthermore, if G has m linearly independent eigenvectors, then the matrix G diagonalizes as PDP^{-1} , where the columns of P are the m eigenvectors of G , and D is a diagonal matrix whose entries are the corresponding eigenvalues of G . Being positive-definite, G has strictly positive eigenvalues, and therefore D contains only zeros and positive real numbers. As a result, we can construct a matrix $\sqrt{D}$, such that $(\sqrt{D})^2 = D$ as follows:

$$\sqrt{D} = \begin{bmatrix} \sqrt{D_{1,1}} & 0 & \cdots & 0 \\ 0 & \sqrt{D_{2,2}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sqrt{D_{m,m}} \end{bmatrix},$$

the entries of which are guaranteed to exist, as every entry of D is greater than or equal to 0. With this in mind, consider the matrix $P\sqrt{D}P^{-1}$. First, we show that $P\sqrt{D}P^{-1}$ is symmetric by showing that it is equal to its transpose. First note

that because G is symmetric, $PDP^{-1} = (P^{-1})^T DP^T$, and therefore $P^{-1} = P^T$. Taking the transpose of $P\sqrt{D}P^{-1}$ results in

$$(P^{-1})^T \sqrt{D} P^T \quad (2)$$

Using the fact that $P^{-1} = P^T$, (1) is equivalent to $P\sqrt{D}P^{-1}$, therefore $P\sqrt{D}P^{-1}$ is a symmetric matrix. Squaring this matrix, we see that it amounts to $P\sqrt{D}P^{-1}P\sqrt{D}P^{-1}$, which is indeed G . Thus, we are done. $\square$

B.B There exists an invertible linear transformation $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $v \cdot_G w = f(v) \cdot f(w)$ for all $v, w \in \mathbb{R}$.

Proof. Let $f(v) = \sqrt{G}v$. First, we show that f is injective, and assume that $f(v) = f(w)$ for some $v, w \in \mathbb{R}^m$. By the definition of f , this is equivalent to $\sqrt{G}v = \sqrt{G}w$. We now show that $\sqrt{G}$ has an inverse, by demonstrating that $(\sqrt{G})^{-1} = P(\sqrt{D})^{-1}P^{-1}$ from problem B.A. Note that since D is a strictly positive diagonal matrix, $\sqrt{D}$ is as well, and is therefore invertible. Then, $P\sqrt{D}P^{-1}P(\sqrt{D})^{-1}P^{-1}$ is indeed I , and we have shown $(\sqrt{G})^{-1}$ to be $P(\sqrt{D})^{-1}P^{-1}$. Left multiplying by $(\sqrt{G})^{-1}$, we see that $v = w$, thus showing that f is injective.

Next, we show that f is surjective, and let y be some element of $\mathbb{R}^m$. Consider $x = (\sqrt{G})^{-1}y$, which is also in $\mathbb{R}^m$. Then, $f(x) = \sqrt{G}(\sqrt{G})^{-1}y$, which is indeed y . Therefore, for any $y \in \mathbb{R}^m$, there exists an $x \in \mathbb{R}^m$ such that $f(x) = y$. As such, we have shown f to be bijective.

We now show that f satisfies the relationship $v \cdot_G w = f(v) \cdot f(w)$ for any $v, w \in \mathbb{R}^m$. By definition, $f(v) \cdot f(w) = (\sqrt{G}v)^T I \sqrt{G}w$. Using rules of transposition, this becomes $v^T \sqrt{G} I \sqrt{G}w$, which is indeed $v^T G w = v \cdot_G w$. Thus, we are done. $\square$